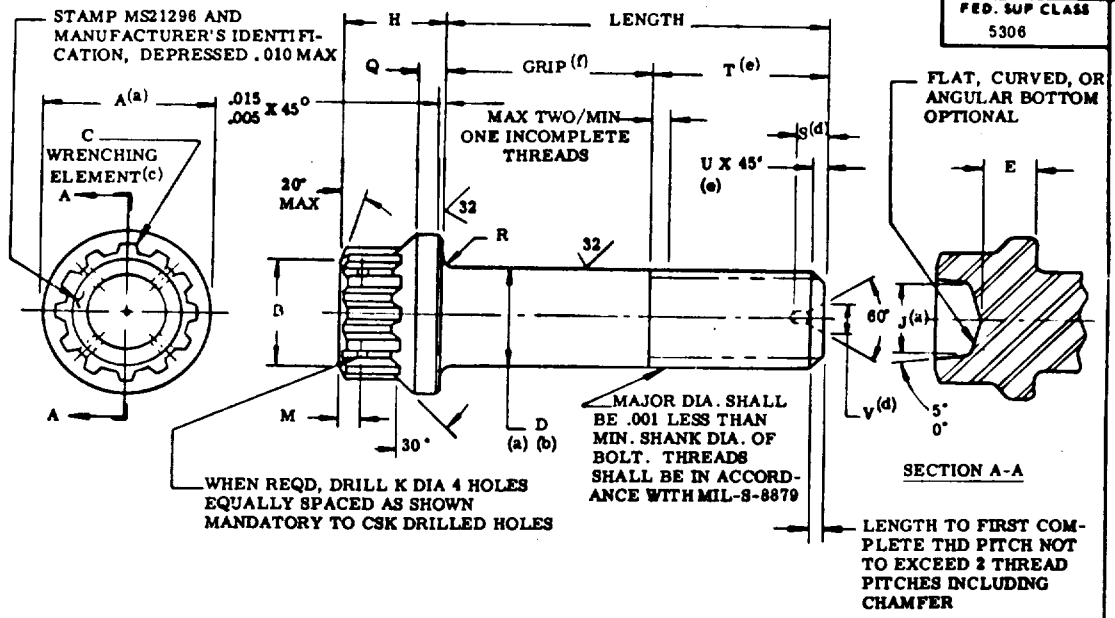
PROJECT NO. 5306-0584

FED. SUP CLASS
5306




NOTES:

(A) 1. MATERIAL: ALLOY STEEL PER AMS 6487.
2. HARDNESS: ROCKWELL C51-54
3. SURFACE TEXTURE IN ACCORDANCE WITH ANSI (ASA) B46.1-1962.
4. PLATING: CADMIUM COATED (VACUUM DEPOSITED): MIL-C-8837, TYPE II, CLASS 2.
5. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±2°.
6. DIMENSIONS TO BE MET AFTER PLATING.
7. DESIGN AND USAGE INFORMATION: THESE BOLTS ARE DESIGNED TO BE USED WITH MS21085 NUTS AND MS21299 WASHERS.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

(A) REVISED

APPROVED 28 JAN 1972 REVISED (A) 6 OCT 1975

| | | |
|---|---|---|
| P.A. NAVY - AS<br>Other Cust<br>USAF-11<br>ARMY-AV | TITLE<br>BOLT, TENSION, STEEL, 260 KSI Ftu, 450°F.,<br>EXTERNAL WRENCHING,<br>SPLINE DRIVE, FLANGED HEAD | MILITARY STANDARD<br>MS21296 |
| PROCUREMENT SPECIFICATION<br>MIL-B-8907 | SUPERSEDES: | SHEET 1 OF 6 |

DD FORM 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PLATE NO. 15320

REVIEWER SYMBOLS: DSA-IS ARMY-AT
USER SYMBOLS: ARMY-MI

"Revise/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS."

This military standard is mandatory for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:31Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS
5306

| DASH NO. | NOMINAL SIZE | THREAD UNJF-3A | A-DIA (a) | | B-DIA | | C (c) WRENCHING ELEMENT | D-DIA. (a) (b) | | DASH NO. |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | MAX | MIN | MAX | MIN | | MAX | MIN | |
| 03 | NO. 10 | .1900-32 | .359 | .349 | .188 | .173 | 8 | .1895 | .1885 | 03 |
| 04 | 1/4 | .2500-28 | .470 | .460 | .250 | .235 | 10 | .2495 | .2485 | 04 |
| 05 | 5/16 | .3125-24 | .585 | .575 | .375 | .360 | 12 | .3120 | .3110 | 05 |
| 06 | 3/8 | .3750-24 | .701 | .691 | .438 | .423 | 14 | .3745 | .3735 | 06 |
| 07 | 7/16 | .4375-20 | .817 | .807 | .500 | .485 | 16 | .4370 | .4360 | 07 |
| 08 | 1/2 | .5000-20 | .932 | .922 | .562 | .547 | 20 | .4995 | .4985 | 08 |
| 09 | 9/16 | .5625-18 | 1.047 | 1.037 | .625 | .610 | 22 | .5615 | .5605 | 09 |
| 10 | 5/8 | .6250-18 | 1.163 | 1.153 | .688 | .673 | 24 | .6240 | .6230 | 10 |
| 12 | 3/4 | .7500-16 | 1.394 | 1.384 | .812 | .797 | 28 | .7490 | .7480 | 12 |
| 14 | 7/8 | .8750-14 | 1.625 | 1.615 | .938 | .923 | 32 | .8740 | .8730 | 14 |
| 16 | 1 | 1.0000-12 | 1.856 | 1.846 | 1.062 | 1.047 | 38 | .9990 | .9980 | 16 |

| DASH NO. | E | H | J (a) +.010 | K | M | Q | | R-(RAD) | | S MAX (d) | T MIN REF (e) | DASH NO. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MIN | ±.010 | -.030 | ±.005 | | MAX | MIN | MAX | MIN | | | |
| 03 | .113 | .205 | .100 | .037 | .062 | .037 | .017 | .022 | .016 | | .438 | 03 |
| 04 | .149 | .250 | .125 | | | .046 | .026 | .028 | .022 | | .520 | 04 |
| 05 | .187 | .312 | .215 | | .070 | .055 | .035 | .035 | .029 | | .634 | 05 |
| 06 | .224 | .337 | .260 | | | .064 | .044 | .043 | .033 | | .719 | 06 |
| 07 | .262 | .393 | .320 | | | .073 | .053 | .049 | .039 | | .833 | 07 |
| 08 | .299 | .450 | .380 | | .094 | .082 | .062 | .055 | .045 | | .913 | 08 |
| 09 | .336 | .505 | .440 | .055 | | .091 | .071 | .061 | .051 | | 1.015 | 09 |
| 10 | .374 | .562 | .500 | | | .100 | .080 | .068 | .058 | | 1.095 | 10 |
| 12 | .449 | .600 | .570 | | | .118 | .098 | .080 | .070 | .200 | 1.293 | 12 |
| 14 | .523 | .700 | .625 | | .125 | .136 | .116 | .093 | .083 | | 1.488 | 14 |
| 16 | .599 | .800 | .740 | | | .154 | .134 | .105 | .095 | .260 | 1.696 | 16 |

APPROVED 28 JAN 1972 REVISED (A) FOR CHGS SEE SHT 1

USER SYMBOLS: ARMY-MI

REVIEWER SYMBOLS: DSA-IS ARMY-AT

"Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS."

This military standard is mandatory for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. NAVY - AS
Other Cust
USAF-11
ARMY-AV

TITLE
BOLT, TENSION, STEEL, 260 KSI Ftu 450°F, EXTERNAL WRENCHING, SPLINE DRIVE, FLANGED HEAD

MILITARY STANDARD
MS21296

PROCUREMENT SPECIFICATION
MIL-B-8907

SUPERSEDES:

SHEET 2 OF 6

DD FORM 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 15320

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:31Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS
5306

| DASH NO. | U REF (e) | V MAX (d) | CENTER DRILL SIZE (d) | BEARING AREA MEAN DIA. SQ. IN. | ULTIMATE TENSILE STRENGTH LB MIN | DOUBLE SHEAR STRENGTH LB MIN | CONCENTRICITY x (a) | CONCENTRICITY z (b) | DASH NO. |
|---|---|---|---|---|---|---|---|---|---|
| 03 | .039 | | | .0517 | 5,640 | 8,840 | .004 | | 03 |
| 04 | .045 | | | .0914 | 10,100 | 15,300 | .005 | | 04 |
| 05 | .052 | | | .1442 | 16,000 | 23,900 | .006 | .0045 | 05 |
| 06 | | | | .2106 | 24,700 | 34,400 | .008 | | 06 |
| 07 | .063 | | | .2896 | 33,500 | 46,900 | .009 | | 07 |
| 08 | | | | .3786 | 44,600 | 61,200 | .010 | | 08 |
| 09 | .069 | | | .4808 | 56,600 | 77,500 | .011 | .0060 | 09 |
| 10 | | | | .5945 | 70,800 | 95,700 | .012 | | 10 |
| 12 | .078 | .187 | 2 | .7512 | 103,000 | 138,000 | .015 | | 12 |
| 14 | .089 | | | 1.1673 | 140,000 | 188,000 | .018 | .0090 | 14 |
| 16 | .104 | .250 | 3 | 1.5465 | 183,000 | 245,000 | .020 | | 16 |

(a) DIMENSIONS A, D AND J SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR X T.I.R.
(b) DIMENSIONS D AND THREAD PITCH DIAMETER SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Z T.I.R.
(c) WRENCHING ELEMENT PER MS33787, DRIVERS PER MIL-W-8982.
(d) ON BOLT SIZES .750 INCH AND LARGER, CENTER DRILLING OF SHANK IS OPTIONAL.
(e) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES ONLY AND ARE NOT INSPECTION OR MANUFACTURING REQUIREMENTS.
(f) GRIP LENGTH OF BOLT SHALL BE MEASURED FROM UNDERSIDE OF HEAD TO THE END OF FULL CYLINDRICAL PORTION OF SHANK.

APPROVED 28 JAN 1972 REVISED (A) FOR CHGS SEE SHT 1

| P.A. NAVY - AS<br>Other Cust<br>USAF-11<br>ARMY-AV | TITLE<br>BOLT, TENSION, STEEL, 260 KSI Ftu, 450°F,<br>EXTERNAL WRENCHING,<br>SPLINE DRIVE, FLANGED HEAD | MILITARY STANDARD<br>MS21296 |
|---|---|---|
| PROCUREMENT SPECIFICATION<br>MIL-B-8907 | SUPERSEDES: | SHEET 3 OF 6 |

DD FORM 672-1 (Coordinated)
PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.
PLATE NO. 15320

USER SYMBOLS: ARMY-MI
REVIEWER SYMBOLS: DSA-IS, ARMY-AT

"Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS."

This military standard is mandatory for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:31Z
Check the source to verify that this is the current version before use.

This military standard is mandatory for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

"Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS."

REVIEWER SYMBOLS:
DSA-IS
ARMY-AT

USER SYMBOLS:
ARMY-MI

P.A. NAVY - AS
Other Cust
USAF-11
ARMY-AV

PROCUREMENT SPECIFICATION
MIL-B-8907

SUPERSEDES:

TITLE
BOLT, TENSION, STEEL, 260 KSI Ftu, 450°F, EXTERNAL WRENCHING, SPLINE DRIVE, FLANGED HEAD

MILITARY STANDARD

MS21296

SHEET 4 OF 6

DD FORM 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 15320

| GRIP DASH NO. | GRIP .010 | DIAMETER DASH NUMBERS 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 12 | 14 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | LENGTH ±.015 | | | | | | | | | | |
| 002 | .125 | .578 | .660 | .774 | .859 | | | | | | | |
| 003 | .188 | .640 | .722 | .836 | .922 | | | | | | | |
| 004 | .250 | .703 | .785 | .899 | .984 | | | | | | | |
| 005 | .312 | .766 | .848 | .962 | 1.046 | 1.160 | | | | | | |
| 006 | .375 | .828 | .910 | 1.024 | 1.109 | 1.223 | 1.303 | | | | | |
| 007 | .438 | .890 | .972 | 1.086 | 1.172 | 1.286 | 1.366 | 1.468 | | | | |
| 008 | .500 | .953 | 1.035 | 1.149 | 1.234 | 1.348 | 1.428 | 1.530 | 1.610 | | | |
| 009 | .562 | 1.016 | 1.098 | 1.212 | 1.296 | 1.410 | 1.490 | 1.592 | 1.672 | | | |
| 010 | .625 | 1.078 | 1.160 | 1.274 | 1.359 | 1.473 | 1.553 | 1.655 | 1.735 | 1.933 | | |
| 011 | .688 | 1.140 | 1.222 | 1.336 | 1.422 | 1.536 | 1.616 | 1.718 | 1.797 | 1.996 | | |
| 012 | .750 | 1.203 | 1.285 | 1.399 | 1.484 | 1.598 | 1.678 | 1.780 | 1.860 | 2.058 | 2.253 | |
| 013 | .812 | 1.266 | 1.348 | 1.462 | 1.546 | 1.660 | 1.740 | 1.842 | 1.922 | 2.121 | 2.316 | |
| 014 | .875 | 1.328 | 1.410 | 1.524 | 1.609 | 1.723 | 1.803 | 1.905 | 1.985 | 2.183 | 2.378 | 2.586 |
| 015 | .938 | 1.390 | 1.472 | 1.586 | 1.672 | 1.786 | 1.866 | 1.968 | 2.047 | 2.246 | 2.441 | 2.648 |
| 016 | 1.000 | 1.453 | 1.535 | 1.649 | 1.734 | 1.848 | 1.928 | 2.030 | 2.110 | 2.308 | 2.503 | 2.711 |
| 017 | 1.062 | 1.516 | 1.598 | 1.712 | 1.796 | 1.910 | 1.990 | 2.092 | 2.172 | 2.371 | 2.566 | 2.774 |
| 018 | 1.125 | 1.578 | 1.660 | 1.774 | 1.859 | 1.973 | 2.053 | 2.155 | 2.235 | 2.433 | 2.628 | 2.836 |
| 019 | 1.188 | 1.640 | 1.722 | 1.836 | 1.922 | 2.036 | 2.116 | 2.218 | 2.297 | 2.496 | 2.691 | 2.898 |
| 020 | 1.250 | 1.703 | 1.785 | 1.899 | 1.984 | 2.098 | 2.178 | 2.280 | 2.360 | 2.558 | 2.753 | 2.961 |
| 021 | 1.312 | 1.766 | 1.848 | 1.962 | 2.046 | 2.160 | 2.240 | 2.342 | 2.422 | 2.621 | 2.816 | 3.024 |
| 022 | 1.375 | 1.828 | 1.910 | 2.024 | 2.109 | 2.223 | 2.303 | 2.405 | 2.485 | 2.683 | 2.878 | 3.086 |
| 023 | 1.438 | 1.890 | 1.972 | 2.086 | 2.172 | 2.286 | 2.366 | 2.468 | 2.547 | 2.746 | 2.941 | 3.148 |
| 024 | 1.500 | 1.953 | 2.035 | 2.149 | 2.234 | 2.348 | 2.428 | 2.530 | 2.610 | 2.808 | 3.003 | 3.211 |
| 025 | 1.562 | 2.016 | 2.098 | 2.212 | 2.296 | 2.410 | 2.490 | 2.592 | 2.672 | 2.871 | 3.066 | 3.274 |
| 026 | 1.625 | 2.078 | 2.160 | 2.274 | 2.359 | 2.473 | 2.553 | 2.655 | 2.735 | 2.933 | 3.128 | 3.336 |
| 027 | 1.688 | 2.140 | 2.222 | 2.336 | 2.422 | 2.536 | 2.616 | 2.718 | 2.797 | 2.996 | 3.191 | 3.398 |
| 028 | 1.750 | 2.203 | 2.285 | 2.399 | 2.484 | 2.598 | 2.678 | 2.780 | 2.860 | 3.058 | 3.253 | 3.461 |
| 029 | 1.812 | 2.266 | 2.348 | 2.462 | 2.546 | 2.660 | 2.740 | 2.842 | 2.922 | 3.121 | 3.316 | 3.524 |
| 030 | 1.875 | 2.328 | 2.410 | 2.524 | 2.609 | 2.723 | 2.803 | 2.905 | 2.985 | 3.183 | 3.378 | 3.586 |
| 031 | 1.938 | 2.390 | 2.472 | 2.586 | 2.672 | 2.786 | 2.866 | 2.968 | 3.047 | 3.246 | 3.441 | 3.648 |
| 032 | 2.000 | 2.453 | 2.535 | 2.649 | 2.734 | 2.848 | 2.928 | 3.030 | 3.110 | 3.308 | 3.503 | 3.711 |
| 033 | 2.062 | 2.516 | 2.598 | 2.712 | 2.796 | 2.910 | 2.990 | 3.092 | 3.172 | 3.371 | 3.566 | 3.774 |
| 034 | 2.125 | 2.578 | 2.660 | 2.774 | 2.859 | 2.973 | 3.053 | 3.155 | 3.235 | 3.433 | 3.628 | 3.836 |
| 035 | 2.188 | 2.640 | 2.722 | 2.836 | 2.922 | 3.036 | 3.116 | 3.218 | 3.297 | 3.496 | 3.691 | 3.898 |

APPROVED 28 JAN 1972 REVISED (A) FOR CHGS SEE SHT 1

FED. SUP CLASS 5306

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:31Z
Check the source to verify that this is the current version before use.

This military standard is mandatory for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

"Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS."

REVIEWER SYMBOLS:
DSA-IS
ARMY-AT

USER SYMBOLS:
ARMY-MI

DD FORM 1426 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 15320

P.A. NAVY - AS
Other Cust
USAF-11
ARMY-AV

PROCUREMENT SPECIFICATION
MIL-B-8907

SUPERSEDES:

TITLE
BOLT, TENSION, STEEL, 260 KSI $F_{tu}$, 450°F, EXTERNAL WRENCHING, SPLINE DRIVE, FLANGED HEAD

MILITARY STANDARD

MS21296

SHEET 5 OF 6

FED. SUP CLASS 5306

| GRIP DASH NO. | GRIP .010 | DIAMETER DASH NUMBERS | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 12 | 14 | 16 |
| | | LENGTH ±.015 | | | | | | | | | | |
| 036 | 2.250 | 2.703 | 2.785 | 2.899 | 2.984 | 3.098 | 3.178 | 3.280 | 3.360 | 3.558 | 3.753 | 3.961 |
| 037 | 2.312 | 2.766 | 2.848 | 2.962 | 3.046 | 3.160 | 3.240 | 3.342 | 3.422 | 3.621 | 3.816 | 4.024 |
| 038 | 2.375 | 2.828 | 2.910 | 3.024 | 3.109 | 3.223 | 3.303 | 3.405 | 3.485 | 3.683 | 3.878 | 4.086 |
| 039 | 2.438 | 2.890 | 2.972 | 3.086 | 3.172 | 3.286 | 3.366 | 3.468 | 3.547 | 3.746 | 3.941 | 4.148 |
| 040 | 2.500 | 2.953 | 3.035 | 3.149 | 3.234 | 3.348 | 3.428 | 3.530 | 3.610 | 3.808 | 4.003 | 4.211 |
| 041 | 2.562 | 3.016 | 3.098 | 3.212 | 3.296 | 3.410 | 3.490 | 3.592 | 3.672 | 3.871 | 4.066 | 4.274 |
| 042 | 2.625 | 3.078 | 3.160 | 3.274 | 3.359 | 3.473 | 3.553 | 3.655 | 3.735 | 3.933 | 4.128 | 4.336 |
| 043 | 2.688 | 3.140 | 3.222 | 3.336 | 3.422 | 3.536 | 3.616 | 3.718 | 3.797 | 3.996 | 4.191 | 4.398 |
| 044 | 2.750 | 3.203 | 3.285 | 3.399 | 3.484 | 3.598 | 3.678 | 3.780 | 3.860 | 4.058 | 4.253 | 4.461 |
| 045 | 2.812 | 3.266 | 3.348 | 3.462 | 3.546 | 3.660 | 3.740 | 3.842 | 3.922 | 4.121 | 4.316 | 4.524 |
| 046 | 2.875 | 3.328 | 3.410 | 3.524 | 3.609 | 3.723 | 3.803 | 3.905 | 3.985 | 4.183 | 4.378 | 4.586 |
| 047 | 2.938 | 3.390 | 3.472 | 3.586 | 3.672 | 3.786 | 3.866 | 3.968 | 4.047 | 4.246 | 4.441 | 4.648 |
| 048 | 3.000 | 3.453 | 3.535 | 3.649 | 3.734 | 3.848 | 3.928 | 4.030 | 4.110 | 4.308 | 4.503 | 4.711 |
| 049 | 3.062 | 3.516 | 3.598 | 3.712 | 3.796 | 3.910 | 3.990 | 4.092 | 4.172 | 4.371 | 4.566 | 4.774 |
| 050 | 3.125 | 3.578 | 3.660 | 3.774 | 3.859 | 3.973 | 4.053 | 4.155 | 4.235 | 4.433 | 4.628 | 4.836 |
| 051 | 3.188 | 3.640 | 3.722 | 3.836 | 3.922 | 4.036 | 4.116 | 4.218 | 4.297 | 4.496 | 4.691 | 4.898 |
| 052 | 3.250 | 3.703 | 3.785 | 3.899 | 3.984 | 4.098 | 4.178 | 4.280 | 4.360 | 4.558 | 4.753 | 4.961 |
| 053 | 3.312 | 3.766 | 3.848 | 3.962 | 4.046 | 4.160 | 4.240 | 4.342 | 4.422 | 4.621 | 4.816 | 5.024 |
| 054 | 3.375 | 3.828 | 3.910 | 4.024 | 4.109 | 4.223 | 4.303 | 4.405 | 4.485 | 4.683 | 4.878 | 5.086 |
| 055 | 3.438 | 3.890 | 3.972 | 4.086 | 4.172 | 4.286 | 4.366 | 4.468 | 4.547 | 4.746 | 4.941 | 5.148 |
| 056 | 3.500 | 3.953 | 4.035 | 4.149 | 4.234 | 4.348 | 4.428 | 4.530 | 4.610 | 4.808 | 5.003 | 5.211 |
| 057 | 3.562 | 4.016 | 4.098 | 4.212 | 4.296 | 4.410 | 4.490 | 4.592 | 4.672 | 4.871 | 5.066 | 5.274 |
| 058 | 3.625 | 4.078 | 4.160 | 4.274 | 4.359 | 4.473 | 4.553 | 4.655 | 4.735 | 4.933 | 5.128 | 5.336 |
| 059 | 3.688 | 4.140 | 4.222 | 4.336 | 4.422 | 4.536 | 4.616 | 4.718 | 4.797 | 4.996 | 5.191 | 5.398 |
| 060 | 3.750 | 4.203 | 4.285 | 4.399 | 4.484 | 4.598 | 4.678 | 4.780 | 4.860 | 5.058 | 5.253 | 5.461 |
| 061 | 3.812 | 4.266 | 4.348 | 4.462 | 4.596 | 4.660 | 4.740 | 4.842 | 4.922 | 5.121 | 5.316 | 5.524 |
| 062 | 3.875 | 4.328 | 4.410 | 4.524 | 4.609 | 4.723 | 4.803 | 4.905 | 4.985 | 5.183 | 5.378 | 5.586 |
| 063 | 3.938 | 4.390 | 4.472 | 4.586 | 4.672 | 4.786 | 4.866 | 4.968 | 5.047 | 5.246 | 5.441 | 5.648 |
| 064 | 4.000 | 4.453 | 4.535 | 4.649 | 4.734 | 4.848 | 4.928 | 5.030 | 5.110 | 5.308 | 5.503 | 5.711 |
| 066 | 4.125 | | 4.660 | 4.774 | 4.859 | 4.973 | 5.053 | 5.155 | 5.235 | 5.433 | 5.628 | 5.836 |
| 068 | 4.250 | | 4.785 | 4.899 | 4.984 | 5.098 | 5.178 | 5.280 | 5.360 | 5.558 | 5.753 | 5.961 |
| 070 | 4.375 | | 4.910 | 5.024 | 5.109 | 5.223 | 5.303 | 5.405 | 5.485 | 5.683 | 5.878 | 6.086 |
| 072 | 4.500 | | 5.035 | 5.149 | 5.234 | 5.348 | 5.428 | 5.530 | 5.610 | 5.808 | 6.003 | 6.211 |

APPROVED 28 JAN 1972 REVISED (A) FOR CHGS SEE SHT 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:31Z
Check the source to verify that this is the current version before use.

This military standard is mandatory for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

"Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS."

REVIEWER SYMBOLS:
DSA-IS
ARMY-AT

USER SYMBOLS:
ARMY-MI

DD FORM 1426 SUP BY 672-1 (Coordinated)

P.A. NAVY - AS
Other Cust
USAF-11
ARMY-AV

PROCUREMENT SPECIFICATION
MIL-B-8907

SUPERSEDES

TITLE
BOLT, TENSION, STEEL, 260 KSI Ftu 450°F, EXTERNAL WRENCHING, SPLINE DRIVE, FLANGED HEAD

MILITARY STANDARD

MS21296

SHEET 6 OF 6

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 15320

| GRIP DASH NO. | GRIP .010 | DIAMETER DASH NUMBERS | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 12 | 14 | 16 |
| | | LENGTH ±.015 | | | | | | | | | | |
| 074 | 4.625 | | 5.160 | 5.274 | 5.359 | 5.473 | 5.553 | 5.655 | 5.735 | 5.933 | 6.128 | 6.336 |
| 076 | 4.750 | | 5.285 | 5.399 | 5.484 | 5.598 | 5.678 | 5.780 | 5.860 | 6.058 | 6.253 | 6.461 |
| 078 | 4.875 | | 5.410 | 5.524 | 5.609 | 5.723 | 5.803 | 5.905 | 5.985 | 6.183 | 6.378 | 6.586 |
| 080 | 5.000 | | 5.535 | 5.649 | 5.734 | 5.848 | 5.928 | 6.030 | 6.110 | 6.308 | 6.503 | 6.711 |
| 082 | 5.125 | | 5.660 | 5.774 | 5.859 | 5.973 | 6.053 | 6.155 | 6.235 | 6.433 | 6.628 | 6.836 |
| 084 | 5.250 | | 5.785 | 5.899 | 5.984 | 6.098 | 6.178 | 6.280 | 6.360 | 6.558 | 6.753 | 6.961 |
| 086 | 5.375 | | 5.910 | 6.024 | 6.109 | 6.223 | 6.303 | 6.405 | 6.485 | 6.683 | 6.878 | 7.086 |
| 088 | 5.500 | | 6.035 | 6.149 | 6.234 | 6.348 | 6.428 | 6.530 | 6.610 | 6.808 | 7.003 | 7.211 |
| 090 | 5.625 | | 6.160 | 6.274 | 6.359 | 6.473 | 6.553 | 6.655 | 6.735 | 6.933 | 7.128 | 7.336 |
| 092 | 5.750 | | 6.285 | 6.399 | 6.484 | 6.598 | 6.678 | 6.780 | 6.860 | 7.058 | 7.253 | 7.461 |
| 094 | 5.875 | | 6.410 | 6.524 | 6.609 | 6.723 | 6.803 | 6.905 | 6.985 | 7.183 | 7.378 | 7.586 |
| 096 | 6.000 | | 6.535 | 6.649 | 6.734 | 6.848 | 6.928 | 7.030 | 7.110 | 7.308 | 7.503 | 7.711 |
| 098 | 6.125 | | | | | | | 7.155 | 7.235 | 7.433 | 7.628 | 7.836 |
| 100 | 6.250 | | | | | | | 7.280 | 7.360 | 7.558 | 7.753 | 7.961 |
| 102 | 6.375 | | | | | | | 7.405 | 7.485 | 7.683 | 7.878 | 8.086 |
| 104 | 6.500 | | | | | | | 7.530 | 7.610 | 7.808 | 8.003 | 8.211 |
| 106 | 6.625 | | | | | | | 7.655 | 7.735 | 7.933 | 8.128 | 8.336 |
| 108 | 6.750 | | | | | | | 7.780 | 7.860 | 8.058 | 8.253 | 8.461 |
| 110 | 6.875 | | | | | | | 7.905 | 7.985 | 8.183 | 8.378 | 8.586 |
| 112 | 7.000 | | | | | | | 8.030 | 8.110 | 8.308 | 8.503 | 8.711 |

DIAMETER DASH NO. AT TOP OF TABLE, PLUS GRIP DASH NO. FIRST COLUMN ON LEFT OF TABLE CONSTITUTES A COMPLETE DASH NO. IDENTIFYING SPECIFIED LENGTHS OF A GIVEN SIZE OF BOLT.

ADD LETTER 'H' BEFORE DASH NO. TO DESIGNATE DRILLED HEAD. FIRST TWO DASH NUMBERS DESIGNATE DIAMETER IN SIXTEENTHS. LAST THREE DASH NUMBERS DESIGNATE GRIP LENGTH IN SIXTEENTHS.

UNASSIGNED DASH NOS. SHALL NOT BE USED.

EXAMPLES OF PART NUMBER: MS21296-04008=1/4-28 BOLT, 1.035 LONG, .500 GRIP, UNDRILLED HEAD.
MS21296H04008=1/4-28 BOLT, 1.035 LONG, .500 GRIP, DRILLED HEAD.
MS21296-05012 5 16-24 BOLT, 1.399 LONG, .750 GRIP, UNDRILLED HEAD.
MS21296H05012=5/16-24 BOLT, 1.399 LONG, .750 GRIP, DRILLED HEAD.

FED. SUP CLASS 5306

APPROVED 28 JAN 1972 REVISED (A) FOR CHGS SEE SHT 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:31Z
Check the source to verify that this is the current version before use.